Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Optimization framework for sheet metal forming based on numerical simulations and surrogate models*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		*Knowledge Services*	
	Medium Category	Small Category	Medium Category	Small Category
	Plastic deformation/Powder	*Sheet Forming Technologies*	*R&D / Engineering Services*	*Production Process Modeling / Simulation*
Concept of Proposed Technology	1. *The proposed technology establishes an integrated optimization framework for sheet metal forming that combines physics-based numerical simulations, microstructure-informed material modeling, and Machine Learning (ML)–based surrogate models to systematically improve challenging forming operation of sheet metal thickness in the range of 100 µm and tooling performance [1,2].* 2. *The core concept is to bridge the gap between material microstructure, process physics, and forming parameters by integrating experimentally validated Finite Element Method (FEM) models into an efficient optimization framework demonstrated on an industrially relevant forming application like bipolar plate design for fuel cells (FC) or water electrolyzer (EL) [3,4].*			

Rationale for Joint R&D Project	*1. High technological complexity and risk in multi-scale and multi-stage FEM-ML integrated framework for process optimization.* *2. Market demand in Korea for rapid, cost-efficient forming process development and to secure long term global competitiveness in optimized and sustainable manufacturing process.* *3. High risk but high impact development of the optimization and prediction like digital twin.* *Fraunhofer IPT's expertise in high efficiency forming process development, press machines and forming tools directly addresses Korea's growing demand for rapid, cost-efficient process development in advanced sheet metal forming. By transferring IPT's simulation based forming technology and production-oriented knowledge, Korean industries can shorten development cycles, reduce manufacturing costs, and strengthen long-term global competitiveness in optimized and sustainable manufacturing processes. By jointly addressing differing market, product, and process requirements in Europe and Korea, the project ensures globally competitive, scalable, and sustainable forming solutions.*
Key Words	*1. Development and experimental validation of microstructure-informed forming models for sheet metal forming, (e.g. Bipolar plate in fuel cells).* *2. Integration of material characterization and forming experiments to link microstructure evolution and anisotropy directly to forming behavior and hardware performance.* *3. Application of ML-based surrogate models to support process and tool design decisions, reducing trial-and-error in tooling and forming setup.* *4. Development of multi-objective process and tool optimization frameworks focusing on formability, dimensional accuracy, tool life, and production stability.* *5. Development of tool wear prediction models using process data and ML to enable preventive maintenance and sustain forming quality in long-term production.*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	*1. Improving the sheet metal forming process in terms of forming quality and geometric accuracy by accurately capturing material behavior (microstructure evolution, anisotropy, and friction).* *2. Optimization of forming tools and process parameters through integrated experimental–numerical approaches with ML used as a supporting tool.* *3. Enhancing tool durability and product quality in continuous production by predicting and mitigating tool wear and forming defects.*			
R&D Target Specifications	*The R&D project targets the achievement of a validated and deployable optimization framework for sheet metal forming, with some of following performance specifications:* *1. accuracy of key forming parameters (like thickness distribution, forming force, springback, etc) within ±10–15% deviation from experimental measurements.* *2. reduction of full-scale FEM computational time by at least 25% through ML-based surrogate models while maintaining solution fidelity.* *3. improvement of forming window robustness and reducing dimensional deviations.* *4. prediction of forming tool life through predictive tool wear modeling.*			

R&D Plan

Working Packages	1st year				2nd year				3rd year			
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
WP1: Material Characterization and Experimental work Key tasks: Mechanical testing, microstructure analysis, friction tests, forming experiments												
WP2: Microstructure-informed FEM Model Development Key tasks: Material modeling, anisotropy, friction, FEM simulation & validation												
WP3: ML-Based Surrogate Model Development Key tasks: Experimental and Simulation data generation and processing, Design of experiments, surrogate model training, accuracy and reliability assessment												
WP4: Process & Tool Optimization Framework Key tasks: Multi-objective optimization, tool and process parameter optimization												
WP5: Tool Wear Modeling & Digital Twin Development Key tasks: Wear data acquisition, ML-based wear prediction												
WP6: Validation, Demonstration & Industrial Assessment Key tasks: Pilot-scale trials, long-term testing, benchmarking												

	Expected Milestones: 1. *Material characterization and baseline forming experiments are completed (Q3)* 2. *Microstructure-informed FEM models are validated (Q5)* 3. *Integrated ML-based surrogate models with FEM are operational (Q7)* 4. *Multi-objective optimization framework is operational (Q9)* 5. *Tool wear prediction models are operational and validated (Q10)* 6. *Fully validated framework on an industrially relevant forming application like bipolar plate design for Hydrogen fuel cell or water electrolyzer is demonstrated (Q12)*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1. *Fraunhofer IPT has high level of experience in precision sheet metal forming and process development and its core technical contributions will include the design and execution of forming experiments, advanced tooling concepts, and process chain development for sheet metal forming, with particular emphasis on high-precision components such as bipolar plates.* 2. *IPT has extensive experience of CAE and material modeling, forming simulations along with validation of the simulation results at the press machines available in-house at IPT.* 3. *Fraunhofer IPT will contribute to the integration of experimental results with numerical simulations and ML-based models for the optimization workflow.*
Expected R&R of Korean SME Partner	1. *Provide practical manufacturing requirements, target product specifications (e.g., bipolar plates or comparable formed components), and performance constraints.* 2. *Participate in parameter studies and support the implementation of optimized solutions within the SME's manufacturing environment.* 3. *Provide product, process and quality data of SME partner or their customer for model and process validation.* 4. *Scaling the optimized process developed by IPT at Korean SME partner.* *A clear and competency-based division of work packages and responsibilities should be established within the consortium, ensuring that each partner contributes according to its specific technical strengths and infrastructure and ensures technology's universal application.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. *Validated ML-FEM based optimization framework for sheet metal forming that integrates microstructure FEM simulations, forming experiments, ML-based surrogate models, and tool wear prediction, enabling accurate, efficient, and robust optimization of forming processes and tooling.* 2. *Access to advanced simulation-driven forming technologies and global best practices, enabling faster process development, reduced trial-and-error, improved product quality, extended tool life, and enhanced global competitiveness in precision and sustainable manufacturing.* 3. *Faster and accurate forming process development workflow which reduces scrap, production cost and time-to-market.* 4. *Enable local and SME industries to compete in precision forming sectors like energy, automotive, aerospace and hydrogen production.*

6. References

(1) M. C. Acar, H. G. Korkmaz, S. Toros, and M. Müller, "Investigation and evaluation of formability of thin metallic bipolar plates by stamping process," Int. J. Hydrog. Energy, vol. 99, pp. 203–211, Jan. 2025, doi: 10.1016/j.ijhydene.2024.12.231.

(2) D. M. Neto, M. C. Oliveira, J. L. Alves, and L. F. Menezes, "Numerical Study on the Formability of Metallic Bipolar Plates for Proton Exchange Membrane (PEM) Fuel Cells," Metals, vol. 9, no. 7, p. 810, Jul. 2019, doi: 10.3390/met9070810.

(3) M. T. Tran, H. Wang, H. W. Lee, and D.-K. Kim, "Crystal plasticity finite element analysis of size effect on the formability of ultra-thin ferritic stainless steel sheet for fuel cell bipolar plate," Int. J. Plast., vol. 154, p. 103298, Jul. 2022, doi: 10.1016/j.ijplas.2022.103298.

(4) Wang, H., Ye,F., & Chen, L. et al. (2017). Sheet metal forming optimization by using surrogate modeling techniques. Chinese Journal of mechanical Engineering, 30, 22–36, (2017). doi: 10.3901/CJME.2016.1020.123

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization